

Social Psychology

Black Uniforms and Aggression in Professional Sports: Failure to Replicate Frank and Gilovich (1988) With Original and Expanded NFL and NHL Data

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Frank and Gilovich (1988) [FG] showed that National Football League (NFL) and National Hockey League (NHL) teams wearing black uniforms were penalized to a greater extent than teams not wearing black in longitudinal archival research exploring 16 seasons. They suggested a cultural association of black and malevolence at the root of their finding. We endeavored to replicate the original study both directly and conceptually as well as employ more recent and expanded data while improving the measurement of the predictor and outcome variables. In contrast to their early findings, we fail to detect the effect in both leagues while conducting multiple analyses. We propose that the original study was deficient due to the impoverished data FG studied, and a more refined investigation failed to show an increase in aggression when NFL and NHL teams play in black uniforms. We do not dispute the basic association but argue, in conclusion, that in a complex, cluttered, repetitive and interactive environment, it is not likely to emerge.

The English language is ubiquitous with expressions connecting the color black and negativity: Black mood. Black sheep. Black Plague. Blackmail. Black market and Blacklist are only a few. This association has been long studied in experimental research and the results are fairly consistent. For example, in a carefully controlled laboratory study, Valdez and Mehrabian (1994) explored emotional reactions to achromatic colors and found that black was considered most arousing and overbearing in comparison to various grays and white. Hupka et al. (1997) used a similar research approach cross-culturally and showed that black was closely associated with anger and fear in German, Mexican, Russian, Polish and American participants.

Frank and Gilovich (1988) [FG] in their influential work, laid the foundations for the association between black uniforms and elevated aggression leading to heightened rule infringement in a non-experimental research setting, specifically in the realm of sports. They argued that earlier research showing the effect (e.g., Williams et al., 1970; Williams & McMurty, 1970) may have been contrived and lacking in ecological validity. FG conducted an impressive series of studies demonstrating first that black uniforms

were perceived as more malevolent in comparison to other colors (after removing the teams' names and logos). They then conducted their foundational longitudinal investigation to establish the effect outside of the laboratory. The researchers affirmed archivally, both in the National Football League (NFL) and the National Hockey League (NHL) [1970-1986 seasons] that black clad uniform teams were penalized to a greater extent than their competitors.

FG subsequently demonstrated the association of black and aggression with experimental work. They orchestrated a similar football play twice. When the defensive team was wearing black, it was judged more aggressive than when wearing white, both by lay people as well as experienced referees. Lastly, Frank and Gilovich showed that ad-hoc teams in black chose more aggressive games to compete in from a given list than those wearing white. Their study, probably due to its originality, comprehensiveness and surprising findings, was abundantly cited in academic journals (more than 830 citations to date) and was also covered by influential popular media outlets such as Sports Illustrated (Boxer, 1989), Wired (Keim, 2012) and NPR (Vedantam, 2012) most likely due to its applied nature and great

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interest in sports research. It was one of the early studies to give rise to a subfield of research exploring uniform colors and performance in sports (Goldschmied et al., 2020).

The first goal of the current investigation was to attempt to conceptually replicate Frank and Gilovich's (1988) NFL archival work but also to address major flaws in the original study. Pertinent to the present research, quickly growing literature in psychology (especially in social psychology) has cast reasonable doubts on the robustness of initially published findings. This literature has compellingly demonstrated that many psychological findings do not hold up in direct and/or conceptual replication attempts and that prestigious journals seem to have favored innovation over rigor in the past (Camerer et al., 2018; OSC, 2015). In addition, it has been stated that psychology as well as sport science suffer from an implausibly high positive result rate. Studies have shown that high-ranking journals in the respective disciplines exceed a positive result rate of 80%. In other words, 80% of the studies that were published were able to confirm their hypotheses (Scheel et al., 2021; Twomey et al., 2021). As a consequence of this finding, it has been argued that it is essential for researchers to invest more effort in a "sound derivation chain between test and theory," for example by spending "more time forming concepts, developing valid measures, establishing the causal relationships between concepts and the functional form of those relationships" (Scheel et al., 2021, p. 744).

Of particular importance to the present research, a comprehensive review on color effects in sports (Goldschmied et al., 2020) has demonstrated high levels of ambiguity and has highlighted several methodological shortcomings in previously published color research in the field of sport. Although the field of sport psychology has begun to acknowledge the "so called replication crisis" in psychology (e.g., Schweizer & Furley, 2016), there are still very few replication attempts in this field. In one of the few in color research Dijkstra and Preenen (2008) failed to replicate Rowe et al. (2005) and outlined various confounding variables in the assignment of uniform colors in judo such as seeding system, prior experience, recovery time that influence more directly who won the competition. In team sports another study (Goldschmied et al., 2022), failed to replicate the previously published finding that soccer teams wearing red performed more successfully than teams wearing other jersey colors. Hence, we consider it a research necessity to scrutinize published research findings within the Zeitgeist of the replication movement and address potential weaknesses of original, innovative findings.

Study I: National Football League Black Uniforms Effects Revaluation

Upon examination, one of the main weaknesses of FG's study (1988) pertains to the outcome variable utilized: yards penalized. First, probably due to the way the data were reported by the league then, the two researchers had to lump together all yardage though, even by their own admission, some penalties were not of an aggressive nature (e.g., offside, illegal motion). Their rebuttal was that aggressive fouls (e.g., unnecessary roughness) were penalized

to a greater magnitude than non-aggressive ones (usually 15 vs. 5 yards, respectively) and thus a cumulative count was overall an adequate measure of aggression. We challenge this assertion on many counts since the frequency with which these two types of penalties occur was not studied. It is possible that non-aggressive based fouls dwarf in frequency aggressive based fouls, which if true calls into question the original assumed relationship between black uniforms and aggression. Another problematic issue with utilizing yards penalized as a measure of aggression is that they are game context dependent, as some fouls are assessed at the spot of the infraction. For example, on December 24th 2011, Baltimore was hosting Cleveland and Joe Flacco the quarterback of the home team passed deep right in the first quarter but defensive back Mike Adams committed 'defensive pass interference' which is not an aggressive infraction but did cost his team dearly, a 60-yard penalty. In line, yards penalized may also not be a good measure of overall aggression as the penalization is sometimes dependent on the team's field position which may cause floor effects (e.g., a defensive team committing an aggressive foul in near proximity to its own goal line, will be forced to retreat a reduced number of yards and not to the official full yardage extent that the penalty usually accords).

Moreover, there seem to be two significant omissions in the original data due to rudimentary reporting by the league, which also hampered the original analysis. First, fouls can be declined by the team subjected to the infraction if the outcome sans the penalty is more favorable to it and thus do not impact the teams' field position. Secondly, when both teams commit fouls on the same play, they offset and thus, again, do not register in a teams' penalized yardage.

Alternatively, with the current data we gleaned, we were able to distinguish between types of infractions and thus filter the outcome variable to generate a more accurate measure of aggression. Due to the problematic nature of yards penalized, which we described above, we chose to use the frequency with which aggressive penalties occurred as our outcome variable. Secondly, our database includes most of the penalties that were called but not enforced (i.e., declined, offsetting). This allowed us to conduct an inclusive analysis, which increased the validity of our study.

Another set of problems in the original study pertained to the predictor variables, namely uniform colors. Even though in about half of the games (i.e. when playing away), teams wear white uniforms, in footnote 6 (p.79), Frank and Gilovich (1988) still included these games in the analysis arguing for "reputation". We contend that this is a difficult argument to make since teams act differently based on locale (Craig et al., 2016) and the black uniform effect was proposed to be taking an effect even when the team was not physically wearing this color. In the current analysis we were able to clearly identify the uniforms worn in each game and thus had precise measurement of the predictor variable. For example, the Oakland Raiders, a team traditionally associated with black uniforms and intimidation (and the most penalized team in the original study), wore their traditional colors almost exclusively but in four home

games actually donned the “opposite” white uniform colors, which we accounted for. We conducted several color comparisons to provide a more nuanced approach to uniform color classification.

In sum, assisted with an influx of game statistics brought about with by technological progress ushered in by the era of big data in sport (Morgulev et al., 2018), we were able to examine this past established research. We first provide a comprehensive outlook of NFL game infractions and then, most importantly, assess if earlier findings of black uniform and elevated aggression stand the test of time and methodological rigor.

Method

All game data were extracted from the R package *nflfastR* (for the 1999–2022 NFL seasons). Infractions were distributed about evenly across the seasons with a minimum of 3.7% of the total in several seasons (2001, 2007, 2008 and 2020) and a maximum of 4.8% in the 2005 season. Similarly, teams contributed about equally in committing infractions both at home (range: 2.7% – 3.4%) and away (range: 2.7% – 3.4%).

The uniform colors teams wore for each game were extracted from the website www.gridiron-uniforms.com/ and matched with data from another website that contains the official team colors, <https://usteamcolors.com/nfl-colors/>.

For each game, we tabulated the date, home team and their uniform colors as well as the away team and uniform colors. Pertaining specifically for the infraction, we recorded which team committed the foul, the type of penalty, whether it was declined or offset with a penalty by the opposing team, the yardage assessed, if it was challenged and the outcome of the challenge.

Overall, we recorded 91,487 infractions for the duration of play studied. The frequencies and percentages of each infraction occurring at or over 1% in the full sample were listed in [Table 1](#) (home vs. away). Most infractions are assessed 5, 10 or 15 yard penalties but some penalties may deviate from this classification as infractions are game context dependent. Overall, there were 38,405 infractions that were assessed five yards (42%), 18,989 were assessed ten yards (20.8%) and 10,025 were assessed fifteen yards (11%). Missing in the number of yards penalized were 11,056 penalties (12.1%) which were either declined (i.e., teams can forgo the penalty if the play resulted in either a score or actually puts them in a better field position), offsetting or constituted a second foul by the same team on the same play.

Our analysis, in departing from the original study, exclusively examined uniform effects when teams were playing at home since the away team overwhelmingly played in white ($n = 5,345$ games, 83% of all away games) while in only 180 games did it play in black and the rest ($n = 893$) in other colors. Aside from the limiting uniform color distribution in making the decision to study only home games, we also relied on past research, which found that home vs. away play made a difference in the behavior and performance of teams. For example, Craig (2016), who also utilized archival data, studied the impact of ambient tempera-

tures on infractions in the NFL and found that “The pattern of results changed when aggressive penalties were analyzed separately between home and visiting teams. The relationship was significant for aggressive penalties committed by the home team, but not for those committed by the visiting team. These diverging results imply that the social context also has a role to play in whether temperature will lead to increased aggression. In the context of football, playing in front of supportive fans may garner a sense of high in-group support, which leads to increased acts of aggression towards opposing teams, and allows temperature to play a significant role in being overly aggressive towards the visiting team.” (p. 209). Indeed, also by our data, home and away teams performed differently (or at least judged as such) as away teams were significantly more likely to commit an infraction ($n = 41,490$) than home teams ($n = 38,941$), $\chi^2(1, N = 80,431) = 80.78, p < .001$.

Craig (2016) also departed greatly from the classification approach taken by the original study, determining that only taunting, face masks, unnecessary roughness, and unsportsmanlike conduct were considered as aggressive penalties. We were broader in our classification and labeled unnecessary roughness (i.e., hitting a player after the whistle blows), roughing the passer, personal fouls (i.e., safety-related infraction), unsportsmanlike conduct (i.e., using abusive or insulting language or gestures to players or officials), taunting (i.e., any flagrant acts or remarks that deride, mock, bait, or embarrass an opponent), roughing the kicker (i.e., contacts the plant leg of the kicker while his kicking leg is still in the air. slides into or contacts the kicker when both of the kicker’s feet are on the ground, and disqualification (similar to unnecessary roughness but flagrant, $n(\text{home}) = 47$) as aggressive infractions. Even with the expanded classification, aggressive penalties amounted to only 13.9% ($n = 5,415$) of all the fouls committed at home which called into question the validity of findings by the original study.

[Table 2](#) shows the ranking of each NFL team (if a franchise relocated it was treated as the same entity, as was done in the original study) based on the ratio of aggressive penalties it committed divided by the games played at home in their traditional uniform colors. The current standings were in contrast to the original table of FG (1988). The Oakland Raiders, which was ranked first for most yards penalized (1970–1986) was now only in 6th place. Pittsburgh, which was ranked third before, was now ranked in 11th place. New Orleans climbed one spot to 3rd. Carolina was now 4th and the last black uniform team, Cincinnati, was well below the mean at the 25th spot. Chicago, which was considered a black uniform team (but plays in midnight blue), was the third least aggressive team in the current ranking. We did not conduct any statistical tests with these data since we considered it not robust due to the small number of teams in the sample but we find the ranking far from impressive in supporting the black-aggression association (for example, blue uniform teams were ranked first, fifth and seventh).

Our dataset while comprehensive was still missing half of the penalties when both teams committed a foul during

Table 1. Infraction frequencies in the NFL (National Football League) based on location and type of violation (1999-2022 seasons)

Infractions by	Home Team		Away Team	
	n	%	n	%
False Start	7,700	19.8	8,193	19.7
Offensive holding	7,673	19.7	7,372	17.8
Defensive Pass Interference	2,569	6.6	3,006	7.2
Defensive Offside	2,395	6.2	2,384	5.7
Defensive Holding	1,916	4.9	2,162	5.2
Unnecessary Roughness	1,909	4.9	2,117	5.1
Illegal Block above the Waist	1,411	3.6	1,404	3.4
Delay of Game	1,396	3.6	1,914	4.6
Roughing the Passer	1,179	3.0	1,181	2.8
Face Mask	1,073	2.8	1,194	2.9
Neutral Zone Infraction	899	2.3	1,180	2.8
Offensive Pass Interference	888	2.3	935	2.3
Illegal Use of Hands	871	2.2	873	2.1
Illegal Contact	760	2.0	833	2.0
Encroachment	725	1.9	865	2.1
Illegal Formation	574	1.5	638	1.5
Unsportsmanlike Conduct	508	1.3	554	1.3
Personal Foul	416	1.1	427	1.0
Intentional Grounding	395	1.0	498	1.2
Total	38,941	100	41,490	100

Note. Included were infractions occurring at or above 1% rate in the full sample. Declined infractions were included. Infractions in bold were classified as aggressive. * Only intentional Face Mask infractions that were assessed 15 yards penalty were included (an accidental Face Mask violation, which is accorded only a 5 yards penalty was counted separately).

the same play known as offsetting fouls ($n = 1476$). Since we had no reason to assume that these missing penalties were not evenly distributed between teams/ uniform colors, we kept the first foul that was listed so in all only 1476 fouls were excluded (these fouls constituted only about 0.016 of all tabulated infractions).

Results

Overall, in the 24 seasons analyzed, we had 1221 games in which the home team wore black (committed 1028 aggressive fouls, mean per game = .84) and 5197 games in non-black uniforms (committed 4387 aggressive fouls, mean per game = .84). A chi-square goodness-of-fit test pertaining to the prevalence of aggressive fouls failed to reach significance, $\chi^2(1, N = 5,415) = .006, p = .94$.

Assessing the relationship somewhat differently, we tested whether aggressive fouls were more likely to occur than non-aggressive ones based on uniform color worn at home. Again, a chi-square test failed to reach significance, $\chi^2(1, N = 38,941) = .114, p = .735$. For black uniformed teams, aggressive penalties constituted 13.8% of all infractions at 1028 out of 7458 infractions while for non-black uniformed, aggressive penalties constituted 13.9% of all infractions at 4387 out of 27096 infractions.

Since these omnibus analyses may have violated the independence-of-errors assumption (Webster et al., 2012) we addressed the problem twofold, first by studying each team

separately and secondly by conducting a generalized estimating equation (GEE) to account for the correlation between multiple observations in the data set (i.e., infringements nested in teams).

As NFL teams in the modern era usually wear several uniform colors throughout the season, we were able to study each separately. Table 3 depicts these within team comparisons. We conducted a series of chi-square tests for the ten teams that wore black uniforms at home (regrettably the Oakland team did not meet the requirements to conduct a robust test). Eight out of the nine failed to reach significance while one did however in the opposite manner than would be predicted by the black uniform-heightened aggression hypothesis. The Baltimore Ravens residuals analysis showed the team to be less aggressive in black and more so in white (with no difference for colored uniform).

Secondly, in order to address the overall nested nature of the data (Ghisletta & Spini, 2004) we utilized GEE. Contrary to the findings of the original study, this regression analysis of 38,941 infractions assessing uniform color effects at home (black vs. non-black) and type of infraction (aggressive vs. non-aggressive) as the outcome variable controlling for same team observations ($n = 32$ levels, Min = 1020, Max = 1435) was not significant, $B = -.013, SE = .0446, 95\% CI = -.1$ to .075, Wald Chi-Square = .080, $p = .777$.

Table 2. Frequency of aggressive penalties by National Football League (NFL) teams (1999-2022 seasons)

Team	Home Color	Home Games	Home Games in Home Color	Aggressive Penalties	Penalties/Game
1. Tennessee	Blue	200	166	171	1.030
2. Dallas	White	199	170	162	0.953
3. NEW ORLEANS	Black	204	156	147	0.942
4. CAROLINA	Black	199	105	98	0.933
5. NY Giants	Blue	199	187	174	0.930
6. OAKLAND*	Black	198	194	180	0.928
7. Jacksonville	Blue	197	98	90	0.918
8. Arizona	Red	198	152	139	0.914
9. Detroit	Blue	193	175	160	0.914
10. Baltimore	Purple	200	149	134	0.899
11. PITTSBURGH	Black	208	201	180	0.895
12. St. Louis*	Blue	202	170	152	0.894
13. Tampa Bay	Red	202	108	95	0.880
14. Minnesota	Purple	200	199	175	0.879
15. Miami	White	195	146	128	0.877
16. San Diego*	Blue	198	152	133	0.875
17. Washington	Red	197	122	105	0.860
18. Seattle	Blue	207	197	168	0.852
19. Philadelphia	Green	211	151	124	0.821
20. Cleveland	Brown	193	142	115	0.810
21. Houston	Blue	175	109	86	0.789
22. Denver	Blue	204	112	88	0.786
23. Buffalo	Blue	198	170	133	0.782
24. San Francisco	Red	202	195	149	0.764
25. CINCINNATI	Black	199	146	109	0.746
26. Atlanta	Red	200	111	82	0.738
27. Kansas City	Red	208	207	151	0.729
28. Green Bay	Green	209	197	140	0.710
29. New England	Blue	222	204	144	0.705
30. Chicago	Blue	201	165	114	0.691
31. NY Jets	Green	194	152	100	0.658
32. Indianapolis	Blue	206	205	132	0.644

Discussion

The goal of the present study was to address calls of replicating innovative original findings (Amir & Sharon, 1990) by conducting a conceptual replication of seminal findings. The current investigation failed to replicate the original FG (1988) study. We could not detect an increase in aggressive penalties when NFL teams were wearing black uniforms in the myriads of contrasts and tests we performed.

The second league explored by the original study to substantiate the effect was the NHL. Together the two sports under examination are known as “contact sports”. They are perceived as highly aggressive and are often lumped together in recent years in the investigation of concussions and head injuries (Prien et al., 2018). Due to the availability of old data, we could attempt both a direct replication of

the original data as well as an updated conceptual replication.

Study II: Direct Replication of National Hockey League Black Uniforms Effects

Method

All game data were extracted from the website hockey-reference.com for the 1971-1986 NHL seasons. The uniform colors teams wore for each game were extracted from the website www.nhluniforms.com. For each game, we tabulated the date, away and home team and their uniform colors. Pertaining specifically to the infraction we recorded both which team and player committed the foul, the type of penalty, and the penalty minutes assessed. We eliminated one team, the California Golden Seals (1971-1976)

Table 3. National Football League (NFL) frequency of aggressive penalties home teams by uniform color (1999-2022 seasons)

Team	Black Uniforms			Colored Uniforms			White Uniforms			Chi Sq.	P-value
	Games	Pen.	Pen./ Game	Games	Pen.	Pen./ Game	Games	Pen.	Pen./ Game		
Pittsburgh	201	180	0.9	6	4	0.67	--	--	--	.34	.56
Oakland	194	180	0.93	--	--	--	4	5	1.25	--	--
New Orleans	156	147	0.94	--	--	--	47	36	0.77	1.25	.26
Cincinnati	146	109	0.75	35	36	1.03	18	14	0.78	2.82	.24
Carolina	105	98	0.93	26	28	1.08	68	60	0.88	.76	.68
Atlanta	69	57	0.83	111	93	0.84	--	--	--	.01	.93
Jacksonville	52	46	0.89	98	90	1.03	46	46	1.00	.37	.83
Baltimore ¹	39	27	0.69	149	134	0.90	12	18	1.50	6.70	.04*
Philadelphia	30	19	0.63	151	125	0.83	29	27	0.93	1.73	.42
Arizona	22	16	0.72	152	139	0.91	24	17	0.71	1.58	.45

Note. Pen = Penalties. 1 – shows sign of increased aggression while wearing white. * = between small and medium effect size, the rest of the contrasts at or below small effect size.

which relocated to Cleveland (Barons 1977-78) and eventually merged with the Minnesota North Stars (a franchise which existed previously throughout). We assumed that the original study did similarly but we could not find any specific methodological reporting. We still kept the penalties incurred by teams playing against this team to maintain consistency across the seasons.

Overall, we recorded 136,631 infractions for the duration of play studied (playoff excluded). At the time, NHL teams were wearing their traditional uniform colors when playing away games in contrast to NFL teams which did so at home. Hence, departing from the original study, we exclusively examined uniform effects when teams were playing away since overwhelmingly the home team played in white (only the LA Kings wore yellow at home for the entire period while the Pittsburgh and Vancouver teams did so prior to their uniform switch). Already mentioned before, Craig (2016) noted differences in the number of penalties when playing at home vs. away in the NFL. We found similarly a great discrepancy as significantly more penalties were assessed for the away ($n = 70,079$) over home teams ($n = 66,582$), $\chi^2(1, N = 136,661) = 89.48, p < .001$.

We also observed major weaknesses with the classification of uniform colors in the original study (i.e., the predictor variable). As opposed to the original analysis of the NFL, here FG study recorded a great skew in the penalty box minutes Z scores. One team, the Philadelphia Flyers, represents an extreme outlier with a score of 2.01 and is labeled as a black uniform team. This team, like the Raiders of the NFL was known for its brutality and earned the moniker of ‘The Broad Street Bullies’ (Jackson, 2004). Once again, with a very small sample size (n teams = 23), one score may represent a great “pull” in support of the hypothesis while the association may be haphazard in nature. Most significantly, the Philadelphia Flyers were improperly labeled as a black uniform team. They played their away games in orange. While the team logo contains black and the uniform also has a black stripe on the sleeve, the team’s identity has long been associated with their primary color, orange. They did have a team logo on their chest which was foremost black as well as a small portion of their sleeves. How-

ever, by and large, the color of the uniforms is orange. Another team, the Chicago Blackhawks was labeled originally in black uniforms but never wore the color (despite their name) during the original study. They instead played in red throughout this period. Correcting for the mislabeling of these two teams (especially the Flyers), beyond its influence over the outcome variable of aggression, exacerbates the potential problem of unequal variances between samples (as black uniform teams now shrink considerably in numbers: one team for the duration and another two for about half of the study time). Having both unequal sample sizes and variances severely affects [statistical power](#) and [Type I error](#) rates (Rusticus & Lovato, 2019).

Results

Most infractions are assessed 2, 5 or 10 minutes in the penalty box but the one exception is when players are assessed a double-minor (twice a 2 minutes penalty now). Overall, there were 113,711 infractions that were assessed 2-minute penalties (83.3%), 16,229 were assessed 5-minute penalties (11.9%) and 5,994 were assessed 10-minute penalties (4.4%). There were 645 double-minor penalties (0.5%) and missing were 82 (0.1%) penalties.

The outcome variable of penalty minutes following an infringement as was practiced in the original study seems as an improved measure of aggression to the one used for the NFL since aggression propelled infraction are proportionately more numerous and are not dependent on field position, however it is still far from adequate since many aggressive penalties are still penalized by mere two minutes (Gee & Leith, 2007) and for some penalties a penalty shot is awarded rather than penalty minutes. An additional problem with studying penalty minutes pertains to the skewness of the data (Webster et al., 2012).

Our analysis for the duration of the original study is generally in line with the original league ranking of FG (1988) [their Table 3]. Utilizing Z scores for penalty minutes as practiced before, the three black uniform teams are at the top third of the league (4-6 place). We did not conduct any statistical tests with these data since we considered it not

robust due to the small number of teams in the sample. However, the congruence in ranking between our data and FG serves only to confirm the fidelity of the original's empirical evidence but not to support the black-aggression hypothesis. The main problem with quasi-experimental designs as such is that one has to account for threats (Webster et al., 2012) and the one immanently present here is the one of history. The NHL became a significantly more violent league from 1971 to 1986. Infractions were rising fast during the period analyzed with 5,828 occurring in the first season [78 regular season games for each team] to more than doubling and peaking during the last season at 13,295 [80 regular season games for each team] underlying a strong correlation, $r(14) = .93, p < .001$. We corroborated this massive increase in penalties with summary data derived independently from hockey-reference.com of average league of penalties per game and obtained a similar result, $r(14) = .85, p < .001$. Plassche et al. (2023) demonstrated this sharp rise in violence for these respective seasons specifically with fighting.

Since two teams switched to black (out of only three that actually played in black) about midway through the period explored when aggression in the NHL peaked, a distorted picture emerges. In the original study, these two teams are compared to the rest of the teams in the league whose data are averaged across the entirety of the seasons studied. This gives the impression that the uniform change to black is the driving force behind aggression elevation while it is an overall league trend. Due to the confluence of statistical threats described above we could not reliably undertake the re-analysis of the original data and opted for an updated conceptual replication.

Discussion

FG contended that players in black either become more aggressive or are perceived as such by the referees. They argued that the association was not a spurious relationship as when teams like the Pittsburgh Penguins switched to black, seemingly the same players demonstrated a rise in penalization (and assumed increase in aggression). A closer scrutiny suggested that a spurious association is more plausible. With the new image brandishing of the Pittsburgh team now in black uniforms (1980–81 season), the club also pursued new aggressive players to add to the roster. In line, the three most penalized players in the season following, Paul Baxter (the single-season leader for the Pittsburgh Penguins with 409 during the 1981–82 season), [Pat Hughes](#) and [Gary Rissling](#) were new additions. Similarly, the Vancouver Canucks, once transitioning to black, had new acquired players contributing 31.6% to their penalties on ice in the subsequent season (together with a new general manager to orchestrate it all).

Simply put, the competitive environment of the 1971–1986 NHL in which this association was studied originally is not suitable in numerous respects to carry out the investigation. First, there were very few black uniform teams in the sample and two in the original study were misclassified (The Philadelphia Flyers and the Chicago Blackhawks) while two played in black after switching for less

than half of the period studied (Pittsburgh Penguins and Vancouver Canucks). In essence, this left one team, the Boston Bruins, to qualify fully (unambiguous black coloring for the duration of the study) out of the full (and still very limited) sample. Quantitatively, there are merely 30 team-seasons in black uniforms vs. 267 team-seasons in non-black. Second, only fouls committed in away games, when teams were actually wearing their colors should have been counted. Home games, when teams wore white (or yellow) should have been excluded, as we argue against the concept of “reputation” assumed by FG. Moreover, it is an especially hard argument to make when teams shift uniform colors from one season to the next or even in the middle of the season. Do they immediately “acquire the reputation” after this abrupt change (after many seasons wearing a different color)? Third, the NHL was a league in great transition during the era with respect for the expression of aggression or even violence with a considerable rise (Plassche et al., 2023). It is questionable that uniform color was the driving force behind the upturn given the overall league trend. The relative rise can be much more wholesomely explained with the changing nature and increased aggression in the league rather than the relative minor switch of uniform colors (which seems to be spurious in nature).

Study III: Replication of National Hockey League Black Uniforms Effects in the Modern Era

While there is no follow-up NFL archival research furthering FG's work (1988), their NHL study was followed by Caldwell and Burger (2011) who explored two seasons of league play when teams were allowed to wear a third black uniform in addition to their regular colored one. Playing against the same opponents in black did not yield an increase in measures such as penalty minutes at home or severe penalties involving roughing and fighting in comparison to when playing not in black. Yet, their sample size was limited ($n = 102$ games). Next, Webster et al. (2012) explored the effect in the most extensive manner. They critiqued Caldwell and Burger arguing that their analysis may have violated the independence-of-errors assumption writing that it “featured focused comparisons, its analyses were flawed because they were done at the game level instead of the team level. That is, they ignored the inherent dependency of pairs of games nested within teams.” (p. 275). Webster et al. in turn demonstrated tremendous support for the ‘black-aggression’ hypothesis in the NHL. Utilizing a comprehensive 25 seasons database (1984–1985 to the 2009–2010 regular seasons, $n = 52,098$ games) of 30 professional hockey teams, the researchers showed, once again, that teams wearing black were penalized more due to aggressive infractions (e.g., fighting, roughing, elbowing, kicking, kneeing, tripping) than teams not wearing black. This difference did not emerge for fouls stemming from mental lapses (e.g., illegal substitutions, too many men on the ice). The researchers also reported a white uniform reverse effect, such that when teams switched to wearing colored jerseys at home, they were penalized more than when they wore white uniforms at home games in previous seasons.

While Webster et al. (2012) was extensive in nature there were a few major limitations to their work as well. The researchers reported that beginning with the 1995–1996 season, the league regularly permitted teams to fashion and wear third jerseys as an alternative to their normal colored jersey in about 10–15 games per season (12–18% of games) and that about 80% of teams choose to go with the “alternate” jersey option that features a secondary team color. However, they seem to have disregarded the difference in this substantial minority of games and classified all games as if teams wore their traditional colors. In addition, during most of the period studied, teams at home played in white uniforms while the visitors wore their traditional colors which was reversed only during the 2003–04 to 2009–10 seasons. Thus, the uniform scheme was mostly opposite than the one studied in the NFL and past research (Craig et al., 2016) has demonstrated differences in aggression manifestation based on locale (i.e., increase in penalization for away teams). Moreover, while the study duration of 25 seasons is impressive, it is vital to consider history as a threat (an alternative explanation) to uniform colors as a driving force behind the effect. The league was ever expanding during this era and looked very different by the end of their investigation. European talent became widespread, the equipment got much larger (especially for goaltenders) and a salary cap was introduced, with the last two making the league more competitive and possibly significantly more aggressive (*The Eras of the NHL*, 2009).

We now extend our investigation to the modern NHL while attempting to address deficiencies in past research which supported the black uniforms-heightened aggression connection.

Method

All game data for the 2016/17–2023/24 seasons were extracted from [NHL.com](https://www.nhl.com), the official website of the league. All in all, 71,513 infractions were recorded. They were distributed about evenly across the seasons with a minimum of 8.4% of the total in the 2020–21 shortened COVID season and a maximum of 14.1% in the 2022–23 season. Similarly, teams contributed about equally in committing infractions both at home (range: 2.8% – 3.8%, with the exception of the newly added franchises of the Vegas Golden Knights at 2.4% and The Seattle Kraken at 1.2%) and away (range: 2.8% – 3.7%, with similar exceptions for the new franchises).

The uniform colors teams wore for each game were extracted as well from the website [www.NHL.com](https://www.nhl.com). We limited the sample size for eight seasons because we wanted to corroborate the archival information with visual inspection of video footage which was undertaken independently by two observers. We limited our investigation to eight seasons because video footage from previous ones was intermittent and unreliable as teams have several jerseys at their disposal and archival data did not correspond with the footage that was available.

For the classification of penalties due to the weaknesses of past research (e.g., incongruency between the duration of penalization and the nature of the infringement and

data skewness), we relied on the extensive work of Gee and Leith (2007) who developed a comprehensive taxonomy of aggressive vs. not aggressive penalties. Their work, in turn, relied on Widmeyer and Birch (1978) as well as Widmeyer and McGuire's (1997). We extended their classifications similarly to infractions that resulted in penalty shots. For the rest of the infringements unclassified by past research which constituted only 9.2% of penalties ($n = 3197$), we used our own discretion (undertaken independently by the authors and then agreed upon mutually). This list for additional aggressive infractions included: Abuse of officials, aggressor, checking from behind, game-misconduct, major, head butting, illegal check to head, match penalty, misconduct and butt ending.

For each game, we tabulated the date, home team and their uniform colors as well as the away team and uniform colors. Pertaining specifically for the infraction, we recorded which team committed the foul, the type of penalty, and the penalty minutes assessed. The frequencies and percentages of each infraction occurring at or over 1% in the full sample were listed in [Table 3](#) with the aggressive penalties emphasized (home vs. away). The absolute majority of infractions are assessed at 2 penalty minutes (88.9%, $n = 64,291$) and are considered minor in nature (1% at 4 minutes are dubbed double-minor). Five-minute penalties constitute 6.5% of the sample ($n = 4,655$) while 10-minute penalties amount to 2.1% ($n = 1,495$). Lastly, 15-minute penalties amount to less than .1% of the overall ($n = 28$).

Similarly to our Study I, as well as in line with past research (e.g., Goldschmied et al., 2022), we studied the association of uniform color on performance exclusively at home. Indeed, the NHL data also showed that home and away teams performed differently (or at least judged as such), with away teams significantly more likely to commit an infraction ($n = 36,966$) than home teams ($n = 34,547$), $\chi^2(1, N = 71,513) = 81.82, p < .001$. This distinction held also when aggressive and non-aggressive fouls were analyzed separately.

[Table 4](#) shows the ranking of each NHL team (if a franchise relocated it was treated as the same entity, as was done in the original study) based on the ratio of aggressive penalties it committed divided by the games played at home in their traditional uniform colors. The current standings are a blend of aggressive teams such as Boston Bruins (ranked 2nd) and Anaheim Ducks (ranked 3rd) as well as moderately to lowly aggressive teams (Los Angeles Kings ranked 17th and Pittsburgh Penguins ranked 21st) and do not seem to support the effect. We did not conduct any statistical tests with these data since we considered it not robust due to the small number of teams in the sample.

Results

Overall, in the 8 seasons analyzed, we had 1,935 games in which the home team wore black (committed 3,293 aggressive fouls) and 7,624 games in non-black uniforms (committed 12,046 aggressive fouls, including white = 142). A chi-square goodness-of-fit test pertaining to the prevalence of aggressive fouls reached significance, $\chi^2(1, N = 15,339) = 14.27, p < .001$.

Table 4. Infraction frequencies in the National Hockey League (NHL) based on location and type of violation (2016/17-2023/4 seasons)

Away			Home		
Penalty	n	%	Penalty	n	%
tripping	6317	17.1	tripping	5791	16.8
hooking	4770	12.9	hooking	4243	12.3
roughing	3551	9.6	roughing	3528	10.2
slashing	3405	9.2	slashing	3140	9.1
interference	3131	8.5	interference	2958	8.6
holding	3115	8.4	holding	2883	8.3
high-sticking	2536	6.9	high-sticking	2519	7.3
fighting	2239	6.1	fighting	2240	6.5
cross-checking	1749	4.7	cross-checking	1532	4.4
delaying-game-puck-over-glass	1074	2.9	delaying-game-puck-over-glass	944	2.7
too-many-men-on-the-ice	876	2.4	too-many-men-on-the-ice	835	2.4
misconduct	557	1.5	misconduct	540	1.6
boarding	496	1.3	boarding	414	1.2
unsportsmanlike-conduct	450	1.2	unsportsmanlike-conduct	390	1.1
interference-goaliekeeper	368	1	holding-the-stick	378	1.1
other	2332	6.3	other	2212	6.4
Total	36,966	100	Total	34,547	100

Similar to study I, we also tested whether aggressive fouls were more likely to occur than non-aggressive ones based on uniform color at home. Again, a chi-square test yielded significance, $\chi^2(1, N = 34,547) = 4.25, p = .39$, Carmer's $V = .011$. For black uniformed teams, aggressive penalties constituted 45.5% of all infractions at 3,293 out of 7,242 infractions while for non-black uniformed, aggressive penalties constituted 44.1% of all infractions at 12,046 out of 27,305 infractions.

However, in the more sensitive test of GEE regression of 34,547 infractions controlling for same team observations ($n = 32$ levels, Min = 413, Max = 1300) assessing uniform color effects at home (black vs. non-black) and type of infraction (aggressive vs. non-aggressive) as the outcome variable contrary to the black-aggression association was not significant, $B = -.055, SE = .037, 95\% CI = -.127$ to $.017$, Wald Chi-Square = 2.22, $df = 1, p = .136$. When the predictor variable of uniform colors was distinguished between black, white and colors, the regression still failed to yield significant results.

As NHL teams in the modern era usually wear several uniform colors throughout the seasons, we were able to study each separately. Table 5 entails team by team analysis for the 21 teams wearing occasionally (but at least in five games) black uniforms. Only three teams (15%) show a black-aggression association and one team, the Boston Bruins, is showing the opposite effect whereupon when wearing white uniforms, the team is significantly more aggressive than in black.

Discussion

There is only scant evidence to support the black-aggression hypothesis in the modern NHL ranks. Similar to our previous investigations, we failed to detect the effect when we conducted a nuanced analysis of the overall data (i.e., when accounting for the nested nature of the data) and also when we conducted a team-by-team analysis separately. Also, we failed to find support for reduced aggression when wearing white uniforms (Webster et al., 2012) as we conducted comparisons within teams across eight seasons while ensuring the overall aggression in the league remained consistent (thus avoiding a history threat).

It is worth noting that NHL uniform colors are far from unitary in quite stark to NFL ones. They are usually the combination of three distinct areas of the chest depicting the team's logo, the surrounding area of the torso and around the logo up to the shoulders (usually the greatest area of the fabric) and the sleeves. Each area usually is in a different color and often they are broken down even further (for example, the Philadelphia Flyers logo is composed of majority black but also of white and orange). This complex scheme does not lend itself to a meticulous study of color influence.

It is also important to note that while the two leagues, NFL and NHL, are often lumped together, they also differ in some important respects as there are specific designated players in the NHL whose main task is to fight deliberately (i.e., enforcers) while the NFL does not hold a similar unofficial designation and acts of aggression like fighting may be planned rather than spontaneous (Goldschmied & Espindola, 2013). The leagues vary in their pace of play, as it is segmented for football play with many stoppages, while

Table 5. Frequency of aggressive penalties by National Hockey League (NHL) teams in their primary uniform color (2016/17-2023/4 seasons)

Team Name	Uniform Color	Games	% of games	Penalties	Penalties per game
1. Florida Panthers	red	304	100	650	2.14
2. Boston Bruins	black	299	97.1	591	1.98
3. Anaheim Ducks	black	241	78.8	471	1.95
4. Nashville Predators	yellow	307	100	584	1.9
5. Tampa Bay Lightning	blue	265	86.3	496	1.87
6. Minnesota Wild	green	306	100	551	1.8
7. Ottawa Senators	red	184	60.1	328	1.78
8. Philadelphia Flyers	orange	230	76.7	410	1.78
9. Montreal Canadiens	red	309	100	541	1.75
10. Washington Capitals	red	244	81.1	419	1.72
11. Calgary Flames	red	269	88.2	456	1.7
12. New York Rangers	blue	306	100	509	1.66
13. Seattle Kraken	blue	121	100	197	1.63
14. Arizona Coyotes	red	158	52.3	254	1.61
15. Colorado Avalanche	red	265	86.6	424	1.6
16. New Jersey Devils	red	267	88.1	428	1.6
17. Los Angeles Kings	black	268	87.6	416	1.55
18. Edmonton Oilers	blue	304	100	467	1.54
19. Detroit Red Wings	red	308	100	469	1.52
20. Winnipeg Jets	blue	267	87	403	1.51
21. Pittsburgh Penguins	black	270	88.8	405	1.5
22. Dallas Stars	green	282	92.5	418	1.48
23. Buffalo Sabres	blue	269	89.4	395	1.47
24. Vancouver Canucks	blue	274	89	401	1.46
25. St. Louis Blues	blue	304	100	441	1.45
26. San Jose Sharks	blue	218	71.7	311	1.43
27. New York Islanders	blue	295	97	409	1.39
28. Columbus Blue Jackets	blue	297	98	404	1.36
29. Toronto Maple Leafs	blue	294	98	396	1.35
30. Vegas Golden Knights	gray	224	84.8	295	1.32
31. Chicago Blackhawks	red	293	97	382	1.3
32. Carolina Hurricanes	red	171	57	186	1.09

it is continuous for ice hockey. Lastly, the NHL is strictly played indoors while the NFL is played both indoors and outdoors. All these aspects may influence visibility and the nature of aggression perpetrated by the players. Again, regardless of these environmental variations, the black uniform effect in a careful analysis did not emerge in the NHL ranks.

General Discussion

FG offered two possible paths linking black uniforms to aggression (after rejecting a spurious association). The first explanation was of a real increase in aggression by the players due to its cultural association with malevolence. In addition, they argued that perception may have also partially also propelled the effect. They contended that the pairing also existed within the referees and the malevolent asso-

ciation led officials to judge black uniformed teams more harshly. Regardless of why the effect was transpiring, the foundation of the association in their research was built upon their archival study of both the NFL and the NHL, which we failed to replicate in both leagues.

In contrast, we argue that the association is most likely spurious. The linking of a certain behavior with a unique uniform color in sports is often plagued by extremely small sample sizes (Goldschmied et al., 2020). For example, Attrill et al. (2008) purportedly established that red uniforms were associated with winning in English soccer utilizing a sample of 68 teams. However, subsequent empirical studies in other leagues and sports (García-Rubio et al., 2011; Goldschmied et al., 2022; Piatti et al., 2012) made the researchers question the connection and postulate that a few strong teams that happened to wear red uniforms drove

Table 6. Frequency of aggressive penalties by National Hockey League (NHL) teams in black, white and other colored uniforms (2016/17-2023/4 seasons)

Team Name	Black		White		Other colors		Chi-square	P-value
	Games	PPG	Games	PPG	Games	PPG		
1. Anaheim Ducks	241	1.95	7	1.43	58	2.12	1.734	0.420
2. Arizona Coyotes	133	1.67	0	0	169	1.61	0.162	0.687
3. Boston Bruins ¹	299	1.98	9	3.11	0	0	5.596	0.018
4. Buffalo Sabres	27	1.93	5	2.4	269	1.47	5.920	0.052
5. Calgary Flames	35	1.57	0	0	270	1.69	2.550	0.613
6. Carolina Hurricanes	121	1.49	0	0	179	1.11	8.070	0.004
7. Chicago Blackhawks	9	1.33	0	0	293	1.3	1.172	0.279
8. Columbus Blue Jackets	6	2.17	0	0	297	1.36	2.779	0.096
9. Dallas Stars	20	0.95	0	0	285	1.48	3.640	0.056
10. Los Angeles Kings	268	1.55	26	1.65	12	1.25	0.879	0.644
11. New Jersey Devils	31	1.65	5	1.8	267	1.6	0.144	0.931
12. New York Islanders	9	1.22	0	0	295	1.39	0.170	0.680
13. Ottawa Senators	121	2.18	0	0	185	1.78	5.971	0.015
14. Philadelphia Flyers	61	1.62	9	1.11	230	1.78	2.765	0.251
15. Pittsburgh Penguins	270	1.5	0	0	34	1.47	0.017	0.895
16. San Jose Sharks	45	1.6	8	.75	251	1.49	3.314	0.191
17. Tampa Bay Lightning	40	2.45	0	0	267	1.89	5.611	0.018
18. Vancouver Canucks	33	1.61	0	0	275	1.47	1.132	0.287
19. Vegas Golden Knights	40	1.35	0	0	224	1.32	0.028	0.867
20. Washington Capitals	57	1.65	0	0	244	1.72	0.126	0.723
21. Winnipeg Jets	40	1.83	0	0	267	1.51	2.235	0.135

1 – shows sign of increased aggression while wearing white

the original supposed association. In regard to aggression and black uniforms, there is anecdotal evidence that the Oakland Raiders (the most aggressive team at the time), through their longtime owner, Al Davis (now deceased), were pursuing especially aggressive players (King, 2004) and simultaneously (but independently) seeking an intimidating image to go along. Ben Davidson, a fearsome former Raider defensive end, was quoted (Boxer, 1989) “The people in control of the Raiders—particularly managing general partner Al Davis—seemed to seek out an aggressive outlaw image.”

The original study was deficient in several respects, mostly due to the times and the missing nature of the data explored by FG (1988) but also because of classification issues. Their investigation failed primarily to distinguish the nature of the violation (aggressive vs. non-aggressive) as well as when indeed teams were wearing their traditional uniform colors and when not. In addition, the original study utilized imprecise and skewed outcome measures. Indeed, a meta-analysis in this domain (Weiß et al., 2022), which also assessed the overall methodological quality of the studies, awarded the original FG study a score below the average and generally lamented the “The results of the quality assessment with the QATSDD revealed a low average quality (the overall average score of the evaluated criteria was 51%, with an SD of 17%). Especially the items A4 (evidence of sample size considered in terms of analysis), A8 (detailed

recruitment data), A9 (statistical assessment of reliability and validity of measurement tools), A15 (evidence of user involvement in design) consistently received low ratings (p. 46).” When we principally addressed these weaknesses and conducted multiple comparisons both within and across teams, the effect was not detected.

Many years subsequent to FG (1988) innovative exploration, Elliot and Maier (2012) offered their broad Color-In-Context theory. They maintained six core tenants: 1) Colors convey specific psychological information. 2) The meaning of the color information is based in human evolved biology as well as learned associations. 3) Perceiving color activates evaluative processes (i.e., “good vs. bad” or “foe vs. friend”). 4) The triggered evaluative processes serve to motivate rudimentary forms of behavior (e.g., approach or avoid). 5) The influence of color on psychological functioning is assumed to be automatic (i.e. without conscious intent or deliberate processing). 6) The meanings of colors and their effects on psychological functioning and behavior are contextual. FG alluded to context in their original study arguing in regards to black clothing: “We do not believe, for example, that the black garments worn by Catholic clergymen or Hassidic Jews make them any more aggressive than their secular peers. Rather, it would seem to be the case that the semantic link between the color black and evil and aggressiveness would be particularly salient in domains that already possess overtones of competition, confronta-

tion, and physical aggression.” (p. 83). While indeed these conditions do exist in the sporting context, they also seem to be doubtful to emerge due to an environment, which is complex, cluttered, repetitive and interactive in nature (Goldschmied et al., 2020).

In their original study, FG (1988) also postulated about other realms of human interaction whereupon uniforms are worn. Specifically, they mentioned police work, suggesting that black uniformed law enforcement might be more likely to be violent towards citizens (and in return would elicit a more forceful reaction from the citizenry). Addressing this proposition, Johnson (2013) explored data obtained from 2000 through 2009 from a sample of urban police departments between departments wearing black and light uniforms and found no differences in rates of citizens killed by the police, complaints of excessive force, rates of officer assaults or unsubstantiated citizen complaints after controlling for community contextual variables (e.g., city population density, violent crime rate, percentage of black residence and geographic region). Just as in sports, basic experimental laboratory research indicated, however, that the public perceived black police uniforms as more aggressive and malevolent (Johnson, 2005).

While archival research in the realm of sports is certainly a worthy field of exploration into aggression, it is not without its limitations. One such major weakness is spelled out persuasively in the NFL 2020 rulebook: “The word “flagrant,” when used here to describe an action by a player, is meant to indicate that the degree of a violation of the rules—usually a personal foul or unnecessary roughness—is extremely objectionable, conspicuous, unnecessary, avoidable, or gratuitous. “Flagrant” in these rules does not necessarily imply malice on the part of the fouling player or an intention to injure an opponent.”. Intent is indeed hard to establish in this type of correlational research, it can only be presumed.

In an interview given post publication (Boxer, 1989), FG (1988) commented on the black uniform and elevated aggression finding: “If the wearing of a black uniform can have such an effect in the laboratory (referring to their experimental work), there is every reason to believe that

it would have even stronger effects on the playing field (or rink)”. We take the opposite view. It seems that while the basic association does exist, the relationship does not arise when the circumstances are multifaceted, unsystematic, and mutually dependent. In other words, the effect does not manifest itself beyond the sterilized confines of the laboratory and thus likely to carry significantly less importance than previously thought.

Contributions

Contributed to conception and design: NG, PF, MR

Contributed to acquisition of data: MR

Contributed to analysis and interpretation of data: NG, MR

Drafted and/or revised the article: NG, PF, MR

Approved the submitted version for publication: NG

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Ethics Statement

This study did not involve testing of human participants.

Competing Interests

No competing interests exist.

Data Accessibility Statement

The data are stored at <https://osf.io/je87n/files/osfstorage>.

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